

Classification Algorithm for Motor Imagery EEG Signals based on Multi-domain Information CNN

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Abstract—In order to solve the problems of low recognition rate of single feature and poor adaptability of motor imagination EEG signals, a multi-input time-frequency-spatial hybrid convolution neural network algorithm is proposed in this paper. Classify and identify the motor imagery EEG signals. Firstly, the time domain signal matrix is obtained by pre-processing and data amplification of EEG signals, and then the time-frequency transform and spatial filtering of EEG signals are carried out by continuous wavelet transform and common space mode, respectively, and the time-frequency matrix and space-time matrix are obtained. Finally, three types of characteristic matrices are input into the parallel hybrid convolution neural network training model, and the final classification results are obtained. The proposed method is verified by using the 2008 BCI competition Datasets2a data set, and the average classification accuracy is 94.81%. The results show that this method can effectively enhance the recognition rate of motor imagination, offering a novel idea and approach for the research on brain-computer interfaces related to motor imagination.

Keywords—brain computer interface, motor imagery, electroencephalography, convolutional neural network

I. INTRODUCTION

The method suggested in this paper contributes to identifying the brain's motor intention, enabling the classification and recognition of motor imagination EEG signals in Brain-Computer Interface systems. It demonstrates significant advantages compared to traditional CNN models with single features. The organizational structure of the paper comprises the following sections:

- **Introduction:** Provides an overview of the research background, current research methods, and problem statements.
- **Related Work:** Offers a detailed introduction to the algorithm model proposed in this paper, explaining specific analysis steps.
- **Results and Discussion:** Validates the algorithm model using publicly available datasets and compares it with current state-of-the-art models.
- **Conclusion:** Summarizes the research content of the entire paper and outlines prospects for future research methods.

II. RELATED WORKS

Brain-computer interface (BCI) establishes the direct communication between human brain and external devices, and realizes the direct control of external devices by analyzing the electrical signals generated by the activity of brain neurons. Electroencephalogram (EEG)[1] is widely used in BCI system because of its low cost and non-invasive characteristics. The study[2] found that when the brain engages in motor imagery(MI), the ipsilateral cerebral cortex exhibits the Event-Related Synchronization (ERS) phenomenon, leading to an increase in the amplitudes of mu and beta rhythms. Simultaneously, the contralateral cerebral cortex undergoes the Event-Related Desynchronization (ERD) phenomenon, resulting in a decrease in the amplitudes of mu and beta rhythms[3]. Extracting high-quality features from EEG is an important research content of MI intention recognition.

Feature extraction is the premise of motion imagination pattern recognition, which mainly includes time domain analysis, time frequency analysis and spatial analysis[4]. Time-frequency domain analysis methods include Power Spectral Density (PSD), Short-Time Fourier Transform (STFT), Wavelet Packet Transform (WPT)[5],etc., which investigate the frequency information of EEG signals as they vary over time. The main method of spatial analysis is Common Spatial Patterns (CSP)[6], which studies the collection of spatial information between channels.

Some researchers employ Convolutional Neural Network (CNN) for EEG signal analysis. Yang[7] proposed a method of automatically selecting regularized FBCSP and convolution neural network based on data evaluation. Xie[8] obtain time-frequency images from EEG signals processed by WPT, using them as inputs for CNN classification. These methods demonstrate high classification recognition rates and robustness, achieving MI-EEG classification and recognition. However, they do not thoroughly address EEG signals from a multi-feature perspective. The study [9] revealed that the CNN network, featuring three branches, outperformed versions with one or two branches.

III. METHODS

The model structure is illustrated in Fig.1. The original EEG data undergo processing and transformation: the time signal matrix is obtained through filtering and sliding time

window amplification of the original EEG signal; the time-frequency matrix is produced by applying continuous wavelet transform to convert the EEG signal into a time-frequency matrix; the spatial information matrix employs the common spatial pattern for spatial filtering and projecting the EEG signal. The high-dimensional features are mapped through the full connection layer, flattened, and connected into the classifier to yield the final classification results.

A. Continuous Wavelet Transform (CWT)

CWT is commonly used to analyze EEG signals. It offers a time-frequency representation by decomposing the signal into wavelets of different scales. In contrast to traditional time-frequency analysis methods, CWT enables signal analysis across a continuous time scale, facilitating the identification of frequency components at different time points. In the analysis of EEG signals, CWT can identify characteristic time-frequency patterns of distinct MI tasks. The expression of the CWT is shown in Equation:

$$CWT(s, \tau) = \int x(t) \psi * \left[\frac{t-\tau}{s} \right] dt \quad (1)$$

Where $x(t)$ is the input signal, s represents the scaling scale of CWT, ψ represents the wavelet basis function and τ is the time offset. Morlet wavelet is a kind of complex wavelet, which is often used to analyze the characteristics of time domain and frequency domain, to deal with oscillation signals and analyze time-frequency. Its time-domain is:

$$\psi(t) = \pi^{-\frac{1}{4}} e^{i\omega_0 t} e^{-\frac{t^2}{2}} \quad (2)$$

The ERD/ERS phenomenon in the brain induced by motor imagination primarily occurs within the 8-30Hz range. Recognizing the dynamic nature of brain activity, the analysis frequency band is extended to 4-36Hz for comprehensive examination. For multi-channel EEG data, channels can be either merged or accumulated. The analysis specifically utilizes the C3, C4, and CZ electrodes relevant to the motor imagination cortex, as depicted in Fig.1(b). Conversely, the accumulation method involves analyzing data from all channels, as shown in Fig. 1(a).

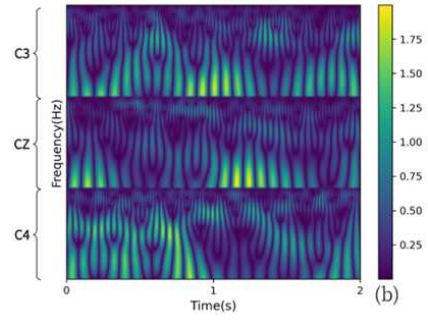
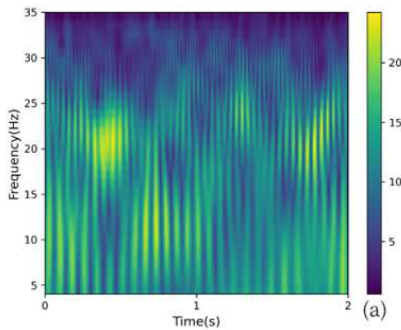


Fig. 1. MI-EEG time-frequency image

B. Common Spatial Pattern (CSP)

CSP is a commonly employed algorithm for extracting features in the processing of EEG signals. Spatial filters are built by relying on two categories of signals. Through the diagonalization of two covariance matrices, it enhances the distinctions between various categories of signals, thereby enhancing the classification performance of various Motor Imagery (MI) tasks.

Where N represents the quantity of electrodes. Subsequently, the covariance matrix R_i of the EEG signal can be formulated as:

$$R_i = \frac{x_i x_i^T}{\text{trace}(x_i x_i^T)} \quad (3)$$

The aggregated spatial covariance matrix can be derived as $R = R_1 + R_2$, R_1 and R_2 signify the mean covariance matrix of the two distinct classes of MI tasks, respectively. The matrix R is subject to decomposition into $R = U \Lambda U^T$, where λ is the characteristic value and U represents the eigenvector of the matrix. The whitening transformation matrix P can be acquired using U and λ . It can be demonstrated that the whitened S_1 and S_2 possess identical eigenvector B , thus obtained.

$$\begin{cases} S_1 = B \lambda_1 B^T \\ S_2 = B \lambda_2 B^T \end{cases}, (\lambda_1 + \lambda_2 = 1) \quad (4)$$

By utilizing the eigenvector B and the whitened EEG signal, the projection matrix W can be calculated.

$$W = B^T P \quad (5)$$

CSP employs a projection transformation to derive spatial features from EEG signals, maximizing differences between different tasks and addressing binary classification problems. In addressing the multi-classification challenge of MI-EEG, the classification problem with n classes is converted into $n(n-1)/2$ binary classification problems. Multiple spatial filters are constructed to project EEG signals, and these filters run in parallel, and it's shown as Fig.2.

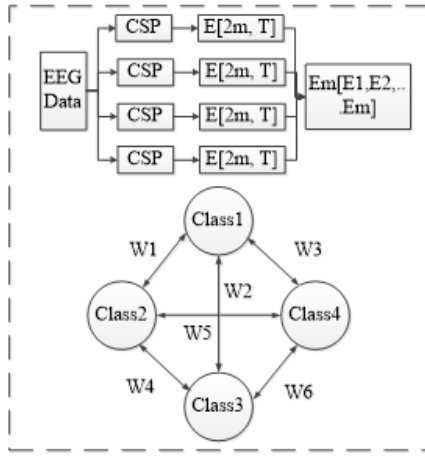


Fig. 2. Single signal acquisition process

C. Proposed CNN Structure

This model comprises three parallel convolutional neural networks that individually analyze the time domain, time-frequency domain, and spatial information of EEG signals.

The input of the first part of the model is the time signal matrix. The convolution kernel sizes of layer 1 and layer 2 of the network are 1×10 and 22×1 respectively, which are used to learn the correlation between channels. A convolution layer with a convolution kernel size of $120'$ is set to learn the time characteristics of the signal. A pooling layer is set up before and after each to reduce complexity and retain significant features.

The input of the second part of the model is to use CWT to obtain two kinds of EEG signal time-frequency matrices with a frequency range of 4-36Hz, which are used to highlight mu rhythm and beta rhythm related to motor imagination. The structures of the two time-frequency matrices are basically similar, and the processing methods in the network are the same. The convolution kernel size of the first layer and the second layer is 5×1 and 10×1 respectively, and the step size is 3×1 and 1×1 , which is used to learn the frequency domain characteristic information in the time-frequency matrix.

The input of the third part of the model is a spatial information matrix composed of spatial characteristics and temporal information after the EEG signal projected by CSP. The first layer pool layer size is set to 8×1 , and the second layer convolution kernel size is set to 6×1 , which corresponds to the number of features in CSP and the number of spatial filters.

After the three segments of the model operate in parallel, a Dropout layer and a fully connected layer are introduced to transform the high-dimensional features from these segments into output results. In the model, the number of neurons in the fully connected layer is set to 128 for feature fusion after multi-input. The parameter of the output layer is set to four, representing the four classification problems of MI-EEG.

D. Model Training

The model training adopts the parallel training of three types of input matrices to obtain the optimal training model. In this study, the Relu activation function is selected due to its fast computation, sparsity, and resistance to saturation. The optimizer choice is the Adam function, which amalgamates the concepts of gradient descent and the momentum method,

effectively adjusting weights to minimize the loss function. For the loss function, sparse classification cross-entropy is utilized to suit the multi-classification MI task. The model parameters are configured as outlined in Table I.

TABLE I. TRAINING PARAMETER SETTING

Parameter name	Numerical value
Training number	40
Size of training data	32
Learning rate	0.001
Activation function	ReLU
optimizer	Adam
loss function	Sparse Categorical Cross entropy
L2 regularization parameter	0.01

IV. RESULTS AND DISCUSSION

A. Database Construction

For this purpose, a public dataset, namely the 2008 BCI Competition IV Data Set 2a [10], was chosen to evaluate the performance of the proposed methodology. The dataset comprises information from 9 subjects, involving 22 scalp electrodes and 3 ophthalmic electrodes. It encompasses four categories of motor imagery tasks: left hand, right hand, feet, and tongue imagination. The data acquisition's experimental paradigm is depicted in Fig.3. Each subject performed 288 motor imagination tasks. The signal's sampling frequency is 250Hz, and the collected signal undergoes bandpass filtering from 0.5 to 100Hz. Additionally, a 50Hz notch filter is applied to eliminate power frequency interference.

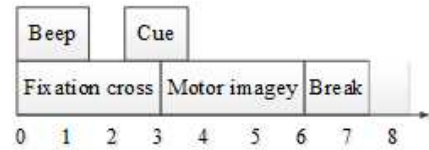


Fig. 3. Single signal acquisition process

In the data preprocessing phase, the Mu and beta rhythms related to EEG are preserved by 4-36Hz band-pass filtering of MI signals. The time segment from 3 seconds to 6 seconds of a single experiment is extracted, and the original signal is segmented accordingly. The segmented data is overlapped and segmented using a sliding time window approach. Through experimentation, it has been verified that setting them to 2 seconds and 0.2 seconds, respectively, facilitates effective data expansion.

B. Experimental Results and Analysis

To assess the model's recognition rates, the three components of the model are iteratively trained to validate the input feature data. The results are then compared with those of the hybrid model, as presented in Table II. The three methods are to use time signal matrix (TS matrix), time-frequency matrix (TF matrix), and spatial information matrix (SP matrix) as CNN input. Subsequently, classification accuracy (Acc) is employed for evaluation.

TABLE II. COMPARISON BETWEEN THE PROPOSED METHOD MODELS

Subject No.	Acc(%)			
	TS matrix	TF matrix	SP matrix	Hybrid model
1	89.88	95.09	85.84	96.40
2	83.24	88.15	84.39	94.66
3	93.64	87.57	84.14	94.95
4	87.57	90.20	79.80	93.27

Subject No.	Acc(%)			
	<i>TS matrix</i>	<i>TF matrix</i>	<i>SP matrix</i>	<i>Hybrid model</i>
5	91.91	86.71	87.28	94.66
6	83.82	93.06	83.28	95.22
7	90.17	93.80	89.88	94.80
8	86.71	88.73	90.46	93.95
9	83.53	91.91	82.08	95.40
Mean	87.38	90.58	85.24	94.81

In contrast among these four methods, the overall performance of the mixed-network model surpasses that of a singular network, showcasing robustness. In addition, we extract the first 288 prediction results from each subject's amplified data, sum the results and confuse the matrix to verify the model results, as shown in fig.4. The figure reveals that the network model with multiple feature inputs achieves higher classification recognition rates than those relying on a single feature, confirming the effectiveness of the multi-input mixed model.

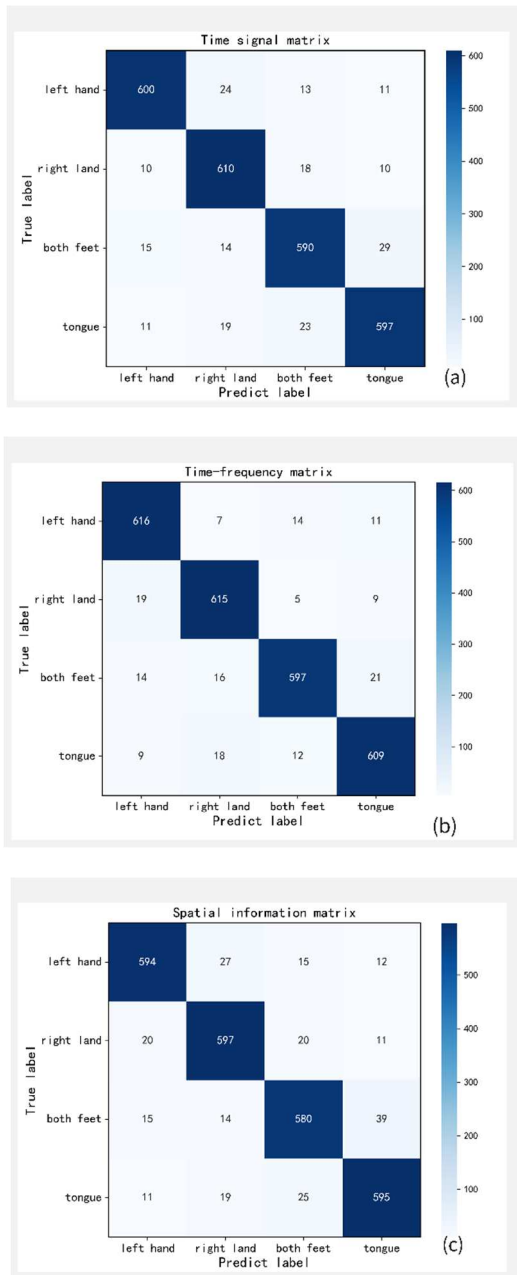


Fig. 4. Confusion matrix of several methods

C. Performance of the Proposed Model

To affirm the superiority of the model proposed in this paper, a comparison was conducted between the classification results of our model and those of several advanced models, as illustrated in Table III.

TABLE III. CLASSIFICATION ACCURACY OF DIFFERENT METHODS

Subject No.	Acc(%)				
	<i>VGGNet [11]</i>	<i>EEGNet [12]</i>	<i>HS-CNN [13]</i>	<i>PSO-CNN [14]</i>	<i>Ours</i>
1	85.36	91.33	90.07	93.30	96.40
2	77.22	72.58	80.28	84.59	94.66
3	92.67	93.14	97.08	91.68	94.95
4	66.12	85.65	89.66	84.55	93.27
5	70.15	83.85	97.04	86.54	94.66
6	71.39	76.98	87.04	76.92	95.22
7	85.73	98.54	92.14	94.03	94.80
8	88.52	89.32	98.51	93.20	93.95
9	72.29	89.99	92.31	92.24	95.40
mean	78.82	86.82	91.57	85.56	94.81

In comparison with various methods, the method outlined in this paper exhibits a higher average classification accuracy compared to the other methods. This underscores the model's capability to learn distinct patterns of electroencephalographic signal activities for different participants during four types of motor imagery states, revealing its level of superiority and effectiveness.

V. CONCLUSIONS

Accordingly, this paper presents a multi-input hybrid CNN algorithm for accurate classification and recognition of MI-EEG, overcoming the limitations of single-domain features. The method processes EEG signals through filtering and sliding time windows to obtain a time signal matrix. CWT generates a time-frequency matrix, while CSP produces a spatial information matrix. These feature matrices are employed as inputs to the hybrid CNN network, undergoing parallel training for motion imagination classification. The achieved excellent classification results underscore the strong representation ability and information richness of our method's feature vectors. Future work encompasses continuous study and refinement of the proposed algorithm, along with the development of an online MI-BCI system based on our algorithm utilizing existing laboratory resources.

AUTHOR CONTRIBUTIONS

Gengyou Xu was responsible for the programming and writing of the paper. Yumiao Ren gave the overall idea. Jieling Jiang participated in the writing of the paper. Fei Yao participated in the algorithm programming. Jiaxin Cao participated in the simulation of the experimental results. The corresponding author is Yumiao Ren.

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